Standard 12
PHYSICS
Part - I

Time: 3.00 Hours Marks: 70

Answer all the questions **15×1=15**

Choose the appropriate answer from the given four alternatives and write the option code and the corresponding answer.

1) Half-lives of two radioactive elements A and B are 20 minutes and 40 minutes respectively. Initially the samples have equal number of nuclei. After 80 minutes the ratio of decayed numbers of A and B nuclei will be:
a) 4 : 1 b) 1 : 4 c) 5 : 4 d) 1 : 16

2) The unit of Planck's constant h is that of
a) work b) force
c) linear momentum d) angular momentum

3) The RMS voltage and frequency of the alternating source V= 230 Sin 314 t is
a) 162.6 V. 50 Hz b) 230 V, 314 Hz c) 230 V, 50 Hz d) 162.6 V, 25 Hz

4) The colour code for a 56 kΩ carbon resistor with tolerance value 5% is
a) Green Blue Orange Gold b) Yellow Green Red Gold
c) Yellow Green Orange Silver d) Green Yellow Red Silver

5) In Joule's law of heating, when R and t are constant, if the H is taken along the y-axis and I^2 along the x-axis, the graph is
a) straight line b) Parabola c) Circle d) ellipse

6) The vertical component of Earths' magnetic field at a place is equal to the horizontal component. What is the value of angle of dip at this place?
a) 30° b) 45° c) 60° d) 90°

7) Fraunhofer lines are an example of ..... specturm.
a) Line emission b) Line absorption
c) Band emission d) Band absorption

8) A dipole (5μc - 5μc) is seperated by 8 mm. The electric potential at a point 20 mm along its equation line will be ..... volt.
a) 9×10^4 b) 18×10^4 c) 18×10^5 d) zero

9) The blue print for making ultra durable synthetic material is mimicked from
a) Lotus leaf b) Morpho butterfly
c) Parrot fish d) Peacock feather

10) The particle which gives mass to protons and neutrons are .....
a) Higgs particle b) Einstein Particle c) Nano particle d) Bulk particle

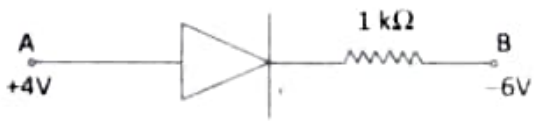
11) A man of height 6 feet wants to buy a plane mirror to see his full image. The minimum heigth of the mirror that he should buy is .....
a) 3 feet b) 6 feet c) 9 feet d) 1.5 feet

12) The nucleus is approximately spherical in shape. Then the surface area of nucleus having mass number A varies as
a) $A^{\frac{2}{3}}$ b) $A^{\frac{4}{3}}$ c) $A^{\frac{1}{3}}$ d) $A^{\frac{5}{3}}$

13) In an electron microscope, the electrons are accelerated by a voltage of 14 KV. If the voltage is changed to 224 KV, then the de Broglie wave length associated with the electrons would
a) Increase by 2 times b) decrease by 2 times
c) decrease by 4 times d) Increase by 4 times

14) Two coherent monochromatic light beams of intensities I and 4 I are superposed. The maximum and minimum possible intesities in the resulting beam are
a) 5 I and I b) 5 I and 3 I c) 9 I and I d) 9 I and 3 I

15) Consider the junction diode is ideal. The value of current flowing through AB is



a) 0 A b) 10^{-2} A c) 10^{-1} A d) 10^{-3} A

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Part - II

Answer any 6 questions. Question No. 22 is compulsory. **6×2=12**

16) Write a short note on electrostatic shielding.
17) Distinguish between drift velocity and mobility.
18) A coil of 200 turns carriers a current of 0 4 A. If the magnetic flux of 4 mWb is linked with the coil, find the inductance of the coil.
19) Why do clouds appear white?
20) List the uses of polaroids.
21) Define work function of a metal. Give its units.
22) The radius of 5^{th} orbit of hydrogen atom is 13.25 A°. Calcuate the de broglie wavelength of the electron orbitting in the 5th orbit?
23) What is meant by skip zone?
24) Mention any two advantages and disadvantages of Roboties.

Part - III

Answer any 6 questions. Question No. 29 is compulsory. **6×3=18**

25) Obtain the expression for energy stored in the parallel plate caption
26) Compute the torque experienced by a magnetic needle in a uniform magnetic field.
27) Mention the various losses in a tranformer.
28) Write down the properties of electromegnetic waves
29) A biconvex lens has radii of curvature 20 cm and 15 cm for the two curved surfaces. The refractor index of the material of the lens is 1.5
    a) What is its focal length?
    b) Will the focal length change if the lens is flipped by the side?
30) Discuss about Nicol prism.
31) Give the construction and working of photo emissive cell.
32) Explain the variation of average binding energy with the mass number using graph and discuss about its features.
33) Prove the following Boolean expressions using the laws and theorms of Boolean algebra

    i) $(A + B)(A + B) = A$

    ii) $A(\bar{A} + B) = AB$

    iii) $(A + B)(A + C) = A + BC$

Part - IV

Answer all the questions. **5×5=25**

34) a) Explain in detail the construction and working of a Van-de-Graaff generator

(OR)

b) Derive the expression for pharse angle between the applied voltatge and current in a series RLC circuit.

35) a) Obtain the equation for bandwidth in Young's double slit experiment

(OR)

b) Desribe the function of a trensistor as an amplifier with the neat circuit diagram. Sketch the input and output wave forms.

36) a) Explain the determination of the internal resistnce of a cell using potentiometer.

(OR)

b) Describe the working of nuclear reactor with block diagram.

37) a) Find the magnetic induction due to a long straigth conductor using Ampere's circuited law.

(OR)

b) What is absorption spectra? Give their types

38) a) Explain and derive the equation for apparent depth.

(OR)

b) Describe briefly Davisson - Germer experiment Which demonstrated the wave nature of electrons.

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Kindly send me your answer keys to us - padasalai.net@gmail.com